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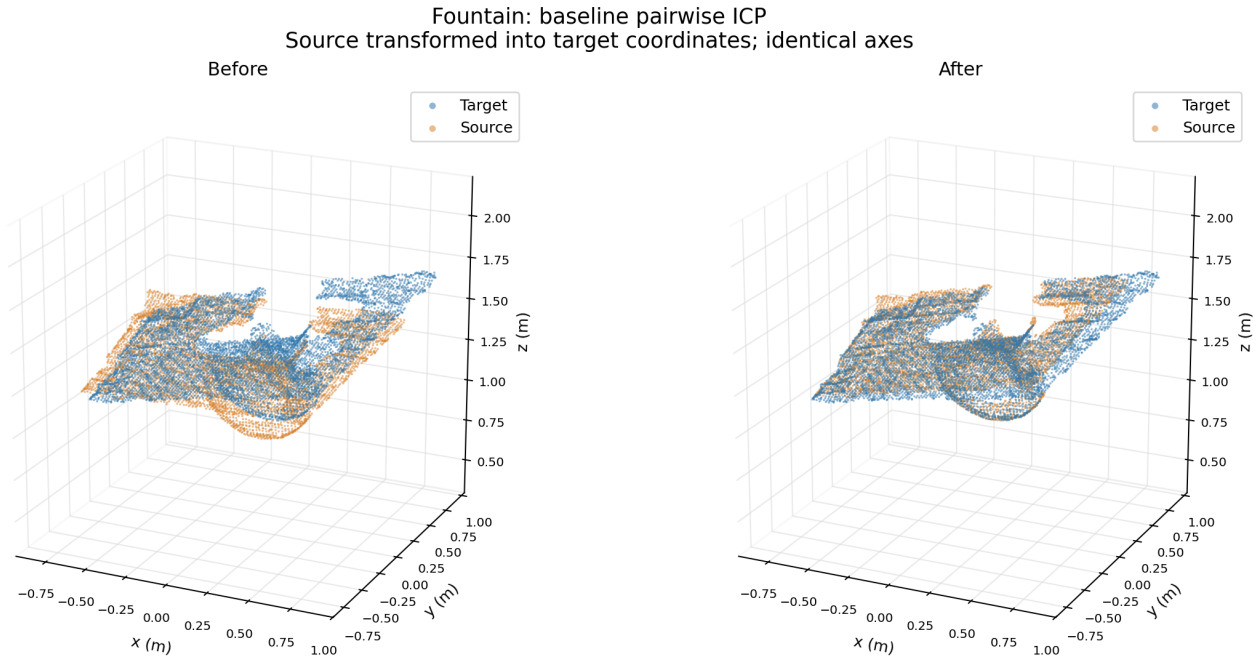
RGB-D Point-Cloud Processing and Registration

Project note | Python, NumPy, Open3D | October 2026

Question and approach

How reliably can two recorded RGB-D views be aligned, and what causes registration to fail? This offline project uses six public Fountain keyframes to examine geometry, preprocessing and pairwise registration.

The pipeline converts depth pixels to coloured 3D points using camera intrinsics, filters invalid depths, applies voxel downsampling and outlier removal, estimates normals, and calls Open3D's point-to-plane ICP solver.



Frame 0 to frame 1: the estimated transform maps source points into target-camera coordinates. Both plots use the same axes.

Recorded findings

Baseline: fixed-evaluation fitness increased from 0.2528 to 0.9658; matched-point RMSE changed from 16.76 to 6.37 mm. Evaluation uses 1 cm voxel clouds and a 3 cm correspondence threshold.

Robustness: 14 experiment records include nine preprocessing/ICP parameter combinations, poor initialization, restricted image coverage, an empty-match check and two additional frame pairs. The poor-initialization case reached only 0.0592 fitness, illustrating a failure case.

Checks: five unit tests cover back-projection, camera-coordinate conventions, a known synthetic rigid transform, agreement with Open3D projection and empty correspondences.

Reproducibility and scope

The source package includes Python modules, pinned dependencies, sample data with provenance and checksums, tests, CSV/JSON results, 4x4 transforms, PLY point clouds and an HTML experiment report. Original run: 23 October 2026. Tests were rerun successfully on 27 October 2026.

Interpretation: fitness is a source-point matching fraction, not accuracy; inlier RMSE is not pose error. The supplied trajectory is an estimated reference, not independent ground truth. This exercise does not demonstrate ROS integration, sensor calibration, live camera capture or deployment on a robot.

Data: Q.-Y. Zhou and V. Koltun, *Color Map Optimization for 3D Reconstruction with Consumer Depth Cameras* (2014), via Open3D SampleFountainRGBDImages. Dataset source | Open3D ICP documentation.